

Zhang Xichen

The Hong Kong Polytechnic University
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Education

Hong Kong Polytechnic University

M.S. Electric Vehicles

Qs world ranking : 54

2024 - 2026

- Graduate with credit
- GPA:3.08/4.00
- Dissertation: Advanced SVPFM for Three-level converter based wireless power transfer system

Nanjing Tech University

B.ENG Electronic Engineering (International cooperation)

Shanghai ranking of Chinese Universities : 89

2020 - 2024

- GPA:83/100 (3.3/4.0)
- Graduate thesis: Design of SVPWM motor control device based on ESP32

Professional Experience

2023 Dataarobtic Co. Ltd., Internship

2024 Sinognss Co. Ltd., Internship

Publications

Under Review

Patent Advanced-SVPFM modulation for Three-level converter based wireless power transfer system 2025

Awards

2023 **Raicom Robot Development Competition**
The Ministry of Industry and Information Technology of China

Provincial Third Prize

2022 **Undergraduate Academic and Technological Forum of Nanjing Tech University**
Nanjing Tech University

Outstanding Award

China Collegiate Computing Contest
CERACU

Second Prize of East China

Qualification

2023 IELTS 6.5

Research Experience

Malicious Domain Defense System Based on Deep Learning

Co-Developer

Nanjing Tech University

2022

- This project won a second prize in the China Collegiate Computing Contest
- Developed a malicious domain name detection and defense system based on deep learning CNN and SDN
- This project was rewarded by Outstanding Work Award as an academic achievement at the Undergraduate Academic Forum of Nanjing Tech University

SCA Motor Development Group Internship

Dataarobotics Co.Ltd

Intern

2023

- Engineered a custom motor communication PCB for robot arms utilizing the BiSS protocol.
- Designed and routed a mixed-signal motor communication PCB, implementing rigorous EMI mitigation and ground isolation to ensure high-speed BiSS protocol reliability.
- Programmed firmware in C for data parsing, and executed hardware debugging using oscilloscopes, heat tests were also conducted..

Hardware & Control Development in Raicom Robot Development Competition

Nanjing Tech University

Independent Developer

2023

- Developed the hardware layout for the robotic motor and power systems, progressing from circuit design to actual physical integration including hardware selection, PCB design through Altium Designer, and debugging.
- Programmed PID feedback loops to dynamically adjust microcontroller PWM outputs, ensuring accurate speed and torque regulation of DC motors.
- This project won the provincial third prize from the Ministry of Industry and Information Technology of China.

Design of SVPWM motor control device based on ESP32

Nanjing Tech University

Graduation Thesis

2023 - 2024

- Engineered a high-precision BLDC motor control system including driver and real-time control feedback from MATLAB simulation to hardware prototype.
- Developed the control system on ESP32 using SVPWM modulation and designed an integrated magnetic encoder PCB for real-time feedback.
- System functions including real-time force feedback, provides real-time motor position feedback via a magnetic encoder integrated into the motor driver PCB.

Advanced-SVPWM modulation for Three-level converter based wireless power transfer system

Hong Kong Polytechnic University

Dissertation

2025

- Derive, design and verified a modulation method for 3-level DC-DC converter for wireless charging as the advanced SVPWM strategy via MATLAB Simulink; derived comprehensive mathematical models to achieve wide-range ZVS for half-bridge flying-capacitor three-level converters in WPT systems.
- Also developed a switching-state-based flying-capacitor voltage balancing method without relying on current measurement, this method is capable of withstanding dynamic step responses.
- Executed a complete simulation-to-hardware developmental cycle: designed the converter into PCB board by Altium Designer, and debugged a physical prototype.
- Utilized SiC MOSFETs with drive and programmed the control strategy to DSP micro-controller hardware experiment. Successfully validated the physical hardware prototype at a 240V operating voltage with frequency over 100k, fully achieving the anticipated performance metrics and system stability.
- Wireless charging system be tested using oscilloscope, DC power supply, electronic load, and power analyzer.
- Key achievements include deriving a closed-form ZVS condition and implementing a sensorless voltage balancing algorithm to realize wide-range soft-switching without circuit modifications, also designed and made an GaN prototype for future improvement.

Relevant Research Skills

- Research experience of DC-DC converter modeling; Modulation strategy design and modeling; Simulation Experiments via MATLAB Simulink
- PCB design via Altium Designer; Experience of IGBT application (SiC & GaN)
- Experience of DSP programming and debugging(TI 28377D) in CCS and ESP32 programming in C
- Testing equipment: oscilloscope, DC power supply, electronic load, power analyzer, Handheld thermal imager